OOES-18: An Open Deterministic Evidence Schema for EU Regulation 2024/3015 and French DELTA IE (H1) Pre-Clearance

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Abstract

Cross-border manufacturing orders expose importers to a structural epistemic gap. A 30% advance telegraphic transfer is released on Day 0 against assertions, while physical production, origin, custody, and documentary truth become observable only at destination arrival. DS-PR412 (SSRN #6661398, DOI 10.2139/ssrn.6661398) is a practice-derived dataset of 412 first orders placed by small founder-led firms in six countries, 2011–2024; 59 orders are coded as adverse (14.32%, about one in seven). The 75-day contract-to-arrival window used here is a reference workflow assumption, not a reported DS-PR412 mean. This paper specifies OOES-18, an open deterministic evidence schema organized as five layers and eighteen slots. Each slot is `UNKNOWN`, `ASSERTED`, or `CORROBORATED` in a bounded Kleene lattice. Two release gates combine state predicates with cross-slot reconciliation. Slots are sealed with RFC 8785 JCS, RFC 5869 HKDF-SHA256, HMAC-SHA256, and AES-256-GCM; attester authenticity is carried by Ed25519 signatures over the payload hash and checked against a relying-party trust store. A salted blind-index protocol detects cross-lender duplicate pledges without sharing tenant keys. The deterministic vector `CASCADE-2026-9F4A2C` halts both gates by design; a one-byte laboratory mutation forces demotion and a `GATE_02` hard stop. We close with the Oracle Boundary Theorem: cryptography proves tenant-side integrity and attester authenticity, not physical ground truth.

1 Empirical Problem and Architectural Invariants

1.1 The 75-day epistemic verification gap

Let t_0 be the instant an importer releases a 30% advance telegraphic transfer, and t_{75} the destination-arrival instant in the reference workflow. This paper models

$$\Delta_{\text{epistemic}} = t_{75} - t_0 = 75 \text{ days}. \tag{1}$$

The 75-day figure sizes the exposure; it is not a mean reported by DS-PR412. During this interval capital is exposed to a dossier that has not been independently corroborated. The factory may be the audited factory; the container may have been stuffed at the declared geofence; the laboratory may have tested the declared batch. At t_0 these are assertions, not established facts. The exposure is not limited to the advance: a documentary credit may cover the balance, a credit insurer may underwrite the receivable, and a bank may finance the shipment. Each participant prices the same unresolved gap, often without a collision-resistant view of whether the shipment is already pledged elsewhere.

DS-PR412 is a two-page, practice-derived working paper covering 412 first manufacturing orders placed by small founder-led firms with factories in Bangladesh, Vietnam, China, India, South Korea, and Thailand across five product categories, 2011–2024. The dataset is author-coded, and an inter-rater reliability review is in progress. Of the 412 orders, 59 are coded as adverse, an incidence of $59/412 = 14.32\%$ (approximately 1 in 7). For OOES-18 purposes, we group the operational failure modes into four families: quality defects against the technical specification; unauthorized shadow subcontracting to an unaudited tier-1 or tier-2 unit; late or forged shipping documents; and TARIC or origin misclassification. This grouping is the specification's, not a classification claimed by DS-PR412. The 14.32% figure is descriptive, not a causal estimate. OOES-18 does not predict these events. It makes the evidence state explicit and forces a deterministic halt when the evidence required to underwrite the event is absent or merely asserted.

1.2 Why legacy layers fail

Post-presentation document examination operates at t_{45}–t_{60}, after the advance is wired and the vessel has sailed. It is retrospective, so a discrepancy cannot recover the Day-0 capital. It is probabilistic: OCR, VLM layout parsing, and LLM extraction return candidate strings with confidence scores, but a score is not an epistemic state. A 0.97 extraction confidence for a container number does not establish that the container was stuffed at the declared factory, nor that it was not pledged to a second lender. Finally, it compares documents to documents, not documents to physical capture events.

Consortium trade-finance blockchains (Marco Polo, we.trade, TradeLens) failed commercially for governance friction, onboarding chicken-and-egg, and misplacement in the stack. A shared ledger can record that a factory asserted a production date; it cannot establish local origin capture. Publishing commercial terms to detect duplicate pledges violates confidentiality; per-bank silos cannot detect cross-bank duplication. OOES-18 is deliberately narrower: it standardizes the evidence object, the epistemic state, the deterministic gate, and a salted collision token, and runs over ordinary APIs and object storage.

1.3 Invariants

- **Determinism.** Every release decision is a pure, total function $\Gamma : \mathcal{D} \to \{\texttt{READY}, \texttt{HARD_STOP}\} \times \mathcal{P}(\mathcal{R}_{\text{halt}})$ of the canonical dossier. No confidence score, fuzzy match, or heuristic weight enters the release path.
- **Three-valued evidence.** Each slot carries a state in $\mathcal{E} = \{\texttt{UNKNOWN}(0) \prec \texttt{ASSERTED}(1) \prec \texttt{CORROBORATED}(2)\}$.
- **Independent corroboration.** A mandatory slot must be `CORROBORATED`: structurally valid, cryptographically sealed, and independently attested by an accredited third party or authority whose attestation is unexpired at t_{ref}.
- **Semantic reconciliation.** Cryptographic validity is necessary but not sufficient. Cross-slot contradictions halt release with enumerated codes.
- **Privacy-preserving duplicate detection.** Lenders detect a repeated shipment identity through a salted blind index, without sharing tenant keys or plaintext commercial terms.

2 Data Model and Epistemic State Machine

2.1 Eighteen slots in five layers

A dossier is $D = (\mathit{tenant}, \mathit{dossier}, t_{\text{ref}}, \mathit{salt}, \{s_1, \ldots, s_{18}\}, B)$. Every slot envelope has seven members: `slot_id`, `layer`, `state`, `payload`, `seal`, `attestation`, and `fault_codes`. The schema closes every object with `additionalProperties: false`. Table 1 lists the slots.

2.2 Kleene lattice and transitions

The state space is a bounded chain:

$$\mathcal{E} = \{\texttt{UNKNOWN}(0) \prec \texttt{ASSERTED}(1) \prec \texttt{CORROBORATED}(2)\}. \tag{2}$$

Table 1: The OOES-18 five-layer, eighteen-slot evidence model. Required payload fields are normative; the JSON Schema closes every payload object.

Slot	Canonical name	Required semantic payload fields
L1.S01	Purchase order	po_number, buyer_eori, supplier_id, incoterms_2020, currency, total_amount, total_units, unit_price, delivery_window_end
L1.S02	Technical pack and BoM	tech_pack_id, sku_code, product_description, fiber_composition, gsm_weight, dimensional_tolerance_pct, bom_hash
L1.S03	Golden sample baseline	sample_seal_tag, physical_sha256_manifest, approved_by_buyer_id, approved_at, colorfastness_baseline
L2.S04	Supplier KYB and factory geo	legal_name, tax_registration_id, country_iso2, factory_lat, factory_lon, beneficial_owners_verified
L2.S05	Registry and forced-labour log	registry_authority, membership_id, rex_number, eu_2024_3015_screening_status, sanctions_clear
L2.S06	Capacity and subcontracting ban	allocated_lines, daily_output_capacity_units, subcontracting_authorized, tier2_mill_tax_ids
L3.S07	Geotagged inline media	media_count, capture_device_pubkey_fp, exif_gps_lat, exif_gps_lon, max_distance_from_s04_meters, merkle_media_root
L3.S08	ISO 2859-1 inline AQL	inspector_agency, iso17020_accreditation_id, aql_standard, sample_size, critical_defects, major_defects, max_allowed_major, result
L3.S09	ISO/IEC 17025 lab and DPP	lab_name, iso17025_accreditation_id, report_number, reach_annex_xvii_compliant, azo_dyes_ppm, measured_cotton_pct, espr_dpp_uid
L4.S10	ISO 17712 bolt seal	seal_serial_number, iso17712_clause_6_compliant, container_number, stuffing_location_type, stuffing_lat, stuffing_lon, sealed_at
L4.S11	Clean on-board B/L	bl_number, scac_code, carrier_name, vessel_name, vessel_imo, port_of_loading_unlocode, port_of_discharge_unlocode, container_number, shipped_on_board_date, gross_weight_kg, carton_count, clean_on_board
L4.S12	Origin proof (certificate or REX)	coo_number, proof_type, origin_regime, preference_programme, issuing_authority, origin_criterion, country_of_origin_iso2, exporter_rex_id, tier2_yarn_origin_iso2, tier3_raw_material_origin_iso2
L4.S13	Settlement instrument	instrument_type, issuing_bank_bic, advising_bank_bic, lc_number, amount, currency, latest_shipment_date, expiry_date
L5.S14	Landed valuation	invoice_number, invoice_date, fob_value_eur, international_freight_eur, insurance_eur, cif_customs_value_eur, valuation_method
L5.S15	TARIC and fiscal schedule	taric_10_code, preference_code, mfn_duty_rate_pct, preferential_duty_rate_pct, calculated_duty_eur, import_vat_rate_pct, calculated_vat_eur
L5.S16	Pre-shipment inspection	psi_agency, certificate_number, verified_carton_count, verified_units_count, verified_gross_weight_kg, verified_seal_serial
L5.S17	GPS geofence chain	origin_geofence_pass, transit_corridor_id, port_gate_in_timestamp, port_unlocode, hardware_tpm_quote_hash
L5.S18	Dossier state root	dossier_merkle_root, slot_digests_map_hash, kernel_version, reviewer_eidas_key_id, signed_at

Meet and join are min and max. UNKNOWN covers missing, unsubmitted, expired, schema-invalid, or cryptographically unauthenticated payloads. It is never coerced to truthy. ASSERTED is a structurally valid, sealed payload uploaded by an interested party with capture provenance. CORROBORATED requires an independent accreditation identifier and a non-expired attestation.

Let $\Sigma_{\text{ev}} = \{\textsc{Capture}, \textsc{Attest}, \textsc{SealOk}, \textsc{SealFail}, \textsc{Expire}\}$. The transition $\delta : \mathcal{E} \times \Sigma_{\text{ev}} \to \mathcal{E}$ is monotone only under cryptographic integrity:

$$\delta(\texttt{UNKNOWN}, \textsc{Capture}) = \texttt{ASSERTED}, \tag{3}$$

$$\delta(\texttt{ASSERTED}, \textsc{Attest}) = \texttt{CORROBORATED}, \tag{4}$$

$$\delta(s, \textsc{SealOk}) = s, \tag{5}$$

$$\delta(s, \textsc{SealFail}) = \texttt{UNKNOWN}, \tag{6}$$

$$\delta(s, \textsc{Expire}) = \texttt{UNKNOWN}. \tag{7}$$

Demotion is a mandatory security override. If the recomputed digest

$$D_{\text{slot}} = \text{HMAC-SHA256}\big(K_{\text{slot}}, \text{UTF8}(\text{JCS}(p))\big) \tag{8}$$

does not equal seal.hmac_sha256, or if attestation.expires_at $< t_{\text{ref}}$, the kernel MUST set the effective state to UNKNOWN and emit ERR_SEAL_INTEGRITY_FAILURE or ERR_ATTESTATION_EXPIRED.

2.3 Gate equations

Let $\sigma(s)$ be the effective state after seal verification. GATE_01_DEPOSIT_PRE_WIRE uses $M_1 = \{S01, \ldots, S06\}$:

$$V_1 = \texttt{READY} \iff \Big[\forall s \in M_1 : \sigma(s) = \texttt{CORROBORATED}\Big] \wedge \Phi_1, \tag{9}$$

where Φ_1 requires S05.sanctions_clear, S05.eu_2024_3015_screening_status="CLEAR", S06.subcontracting_authorized=false, and S06.daily_output_capacity_units>0. Otherwise the gate returns HARD_STOP with the exact uncorroborated slots and semantic codes.

GATE_02_BALANCE_AND_PRE_CLEARANCE uses the core set $M_2^{\text{core}} = \{S01, S02, S03, S04, S07, S08, S09, S10\}$ and the reconciliation set $M_2^{\text{rec}} = \{S05, S11, S12, S14, S15, S16\}$ (14 slots). It returns READY iff

$$\Big[\forall s \in M_2^{\text{core}} \cup M_2^{\text{rec}} : \sigma(s) = \texttt{CORROBORATED}\Big] \wedge \Phi_2, \tag{10}$$

with

$$\begin{aligned}
\Phi_2 :\ & \text{haversine}(S04, S07) \le 500 \text{ m}, \\
& \text{if S10.stuffing_location_type="FACTORY"}: \\
& \quad \text{haversine}(S04, S10) \le 500 \text{ m}, \\
& \text{S08.result="PASS"},\ \text{S08.critical_defects=0}, \\
& \text{S08.major_defects} \le \text{S08.max_allowed_major}, \\
& \text{S09.reach_annex_xvii_compliant=true}, \\
& \text{S09.azo_dyes_ppm<30}, \\
& \big|\text{S09.measured_cotton_pct} - 100\big| \le 3.0, \\
& \text{S10.container_number} = \text{S11.container_number}, \\
& \text{S11.bl_number starts with S11.scac_code}, \\
& \text{S10.container_number starts with S11.scac_code}, \\
& \text{S11.clean_on_board=true}, \\
& \text{S04.country_iso2} = \text{S12.country_of_origin_iso2}, \\
& \text{if S12.origin_criterion="WO"}: \text{ all inputs domestic}, \\
& \text{if S12.proof_type="REX_STATEMENT"}: \\
& \quad \text{S12.origin_regime="PREFERENTIAL"}, \text{S05.rex} = \text{S12.rex}, \\
& \text{if S12.proof_type="CHAMBER_CERTIFICATE"}: \\
& \quad \text{S12.origin_regime="NON_PREFERENTIAL"}, \\
& \big|\text{S14.fob} + \text{S14.freight} + \text{S14.insurance} \\
& \quad - \text{S14.cif}\big| \le 0.01, \\
& \text{S15.taric_10_code is 10 digits.}
\end{aligned} \tag{11}$$

Port-side stuffing is legitimate: when S10.stuffing_location_type="PORT", the factory geofence

is not evaluated. A semantic check evaluates only when every source slot is CORROBORATED; otherwise it returns NOT_EVALUATED and cannot contribute a pass. This prevents a gate from relying on data it does not require. Every failure is reported as a structured \{code, slot_id, detail\} tuple. The halt vocabulary includes ERR_SLOT_ASSERTED, ERR_SLOT_UNKNOWN, ERR_SEAL_INTEGRITY_FAILURE, ERR_ATTESTATION_EXPIRED, ERR_SEMANTIC_GEOFENCE_DISTANCE, ERR_SEMANTIC_CONTAINER_MISMATCH, ERR_SEMANTIC_CIF_ARITHMETIC, and ERR_SEMANTIC_TARIC_FORMAT.

3 Cryptographic Protocol and Blind Index

3.1 RFC 8785 canonicalization

For payload p, JCS(p) sorts object keys by UTF-16 code units, removes insignificant whitespace, preserves array order, escapes strings per ECMA-262, and serializes numbers with ECMAScript Number::toString. It rejects NaN, Infinity, undefined, bigint, functions, non-plain objects, and cycles. The canonical hash is

$$H_{\mathrm{JCS}} = \mathrm{SHA256}\big(\mathrm{UTF8}(\mathrm{JCS}(p))\big). \tag{12}$$

3.2 HKDF-SHA256 slot isolation

The input keying material is a 32-byte tenant master secret; the salt is a 16-byte dossier salt. RFC 5869 derives

$$K_{\mathrm{slot}} = \mathrm{HKDF\text{-}SHA256}(\mathit{IKM}, \mathit{salt}, \mathit{info}, 32), \tag{13}$$

$$\mathit{info} = \texttt{"OOES-18|v1|"} \parallel \mathit{tenant} \parallel \texttt{"|"} \parallel \mathit{dossier} \parallel \texttt{"|"} \parallel \mathit{slot}. \tag{14}$$

Changing any identifier by one character yields an uncorrelated key. Tenant, dossier, and slot domains are therefore cryptographically separated even under one master secret. The deterministic slot digest is

$$D_{\mathrm{slot}} = \mathrm{HMAC\text{-}SHA256}\big(K_{\mathrm{slot}}, \mathrm{UTF8}(\mathrm{JCS}(p))\big). \tag{15}$$

3.3 Authenticated sealing

Encryption at rest uses AES-256-GCM with a 96-bit IV and

$$\mathit{AAD} = \mathrm{UTF8}(\mathit{dossier} \parallel \texttt{"|"} \parallel \mathit{slot} \parallel \texttt{"|"} \parallel \mathit{state}), \tag{16}$$

$$(C, T) = \mathrm{AES\text{-}256\text{-}GCM}\big(K_{\mathrm{slot}}, \mathit{IV}, \mathit{AAD}, \mathrm{UTF8}(\mathrm{JCS}(p))\big). \tag{17}$$

The seal stores hkdf_salt, jcs_sha256, hmac_sha256, aes_gcm_iv, aes_gcm_ciphertext, aes_gcm_tag, and sealed_at. The verifier re-derives K_{slot}, recomputes H_{JCS} and D_{slot}, decrypts the ciphertext, and compares plaintext to the canonical form. Any mismatch demotes the slot.

3.4 Ed25519 attestation signatures

The tenant HMAC is a tenant-side integrity control, not independent evidence: any holder of the master secret can recompute it. Attestation authenticity is carried by a separate Ed25519 signature over the payload hash. Each attestation carries signature_algorithm="Ed25519", a 32-byte attester_public_key_hex, the payload hash payload_jcs_sha256, and a 64-byte signature_hex. The signed message is

$$M = \mathrm{JCS}\big(\{\, \mathit{id},\ h,\ \tau,\ a,\ \alpha,\ t_a,\ t_e,\ \mathrm{alg},\ \mathit{pk} \,\}\big), \tag{18}$$

where id is the slot identifier, h is payload_jcs_sha256, τ is the attester type, a is the authority name, α is the accreditation identifier, t_a and t_e are the attested and expiry timestamps, alg is "Ed25519", and pk is the attester public key. The verifier checks $\mathrm{Ed25519Verify}(\mathit{pk}, M, \sigma) = 1$ and, for a CORROBORATED slot, requires pk to be present in the relying party's trust store for the stated authority, accreditation identifier, and slot scope at t_{ref}. A valid signature from an untrusted key yields ERR_ATTESTATION_UNTRUSTED; a payload hash mismatch yields ERR_ATTESTATION_PAYLOAD_MISMATCH. This gives a bank a verification path that does not require the tenant master secret: recompute the payload hash, compare it to payload_jcs_sha256, and verify the Ed25519 signature with the attester's trusted key.

3.5 Dossier state root

Let $\mathit{id}_i, \sigma_i, D_i$ be the identifier, effective state, and authenticated digest of slot S_i, with $D_i = \epsilon$ for an unsealed slot. The root over $S01$–$S17$ is

$$R = \mathrm{SHA256}\Big(\mathrm{UTF8}\big(\mathop{\Big\|}_{i=1}^{17} (\mathit{id}_i \parallel \texttt{":"} \parallel \sigma_i \parallel \texttt{":"} \parallel D_i \parallel \backslash n)\big)\Big), \tag{19}$$

where $\|$ denotes direct concatenation and $\backslash n$ is the literal newline byte. The slot digest map hash is $\mathrm{SHA256}(\mathrm{UTF8}(\mathrm{JCS}(\{\mathit{id}_i \mapsto D_i\})))$. L5.S18 carries the root once a reviewer signs. In the default vector S18 is UNKNOWN, so the root is computed but not countersigned.

3.6 Salted blind index

Shipment identity normalizes to

$$\begin{aligned} N = {} & \mathrm{Upper}(\mathrm{Trim}(\mathit{BL})) \parallel \texttt{"|"} \parallel \mathrm{Compact}(\mathit{CTR}) \\ & \parallel \texttt{"|"} \parallel \mathrm{Compact}(\mathit{TIN}) \parallel \texttt{"|"} \parallel \mathrm{ISO}(\mathit{DATE}), \end{aligned} \tag{20}$$

where Compact removes all whitespace and hyphens and uppercases the result, and ISO(DATE) is YYYY-MM-DD. The collision token is

$$\mathit{blind_index} = \mathrm{HMAC\text{-}SHA256}(\mathit{NetworkSalt}, \mathrm{UTF8}(N)). \tag{21}$$

Stripping separators is security-relevant: MSCU-582910-4 and MSCU5829104, or BD-BIN-004918273-0101 and BDBIN0049182730101, must collide. Two banks sharing only the network salt compute the same token without exchanging tenant keys, invoice value, or commercial terms.

Threat model. A lender without another tenant's master secret cannot derive its slot keys or plaintext. A shared network salt reveals only that two submissions share a normalized tuple, not the amounts or currencies. HMAC-SHA256 with a 256-bit salt resists preimage and second-preimage attacks. The tuple is structured, not uniformly random: B/L numbers follow carrier patterns and container/tax identifiers are partly predictable. The salt must be access-controlled, queries rate-limited, and salts rotated; the blind index is a duplicate-detection signal, not a privacy guarantee against a salt holder. A collision is a review signal, not a fraud verdict.

3.7 Reference vector

The pinned fixtures are

$$\begin{aligned} \mathit{master} &= \texttt{0102...1f20} \quad (32 \text{ bytes}), \\ \mathit{hkdf_salt} &= \texttt{a1b2...8f90} \quad (16 \text{ bytes}), \\ \mathit{network_salt} &= \texttt{f0e1...1e0f} \quad (32 \text{ bytes}). \end{aligned}$$

The vector CASCADE-2026-9F4A2C carries a Dhaka (BDCGP) to Le Havre (FRLEH) shipment of 12,000 jackets at EUR 12.00, FOB EUR 144,000.00, freight EUR 4,200.00, insurance EUR 300.00, CIF EUR 148,500.00, TARIC 6203321000, B/L MSCU260984112, container MSCU5829104. It has 8 CORROBORATED, 4 ASSERTED, and 6 UNKNOWN slots. Its root is d2eb3498...a67893 and its blind index is 3eea0224...1b4125. Both gates hard-stop on the asserted and unknown slots. Full corroboration of S05, S06, S11, S12, S14, S15, and S16 transitions both gates to READY; a container, stuffing-geofence, origin-criterion, carrier, or CIF contradiction returns them to HARD_STOP even with valid seals.

4 Institutional Outputs and the Oracle Boundary

4.1 View A: pre-wire treasury gate

The ERP receives pre_wire_gate: verdict, the 30% advance amount, the mandatory slot set, the uncorroborated slots, and every halt reason. The ERP MUST NOT release the Day-0 wire on HARD_STOP. The amount is reported for treasury reconciliation; it is not an authorization.

4.2 View B: UCP 600 and ISBP 821 matrix

The lc_mt700_check view maps the same evidence to a pre-presentation discrepancy matrix (Table 2). It runs before doc-

Table 2: View B: UCP 600 / ISBP 821 mapping.

Reference	OOES-18 check and sources
UCP 600 14(d)	Data must not conflict across documents; S04, S05, S10, S11, S12
UCP 600 18; ISBP 821 C1	Invoice issued by beneficiary, in credit currency, reconcilable to PO; S01, S14
UCP 600 20; ISBP 821 E1	Named vessel, shipped on board, clean B/L; S10, S11
UCP 600 27	No defective-condition clause; S11
UCP 600 30	5% quantity tolerance; S01, S16
ISBP 821 L1	Origin, REX, and consignee consistency; S04, S05, S12

Table 3: View C: French DGDDI DELTA IE H1 data elements.

DE	Element	OOES-18 source
11 01	Declaration type H1	S15
12 03	Supporting documents	S01, S09, S11, S12, S14; codes N380, N705, U164/N865
13 01	Exporter identity	S04
13 04	Importer EORI	S01
14 03	Customs value (UCC 70/71)	S14
16 08 / 16 09	Origin, preferential and non-preferential	S12
18 09	10-digit TARIC	S15
19 01 / 19 07	Container and ISO 17712 seal	S10, S11

uments reach the negotiating bank, not at post-presentation OCR time.

CIF customs valuation follows UCC Articles 70 and 71, not ISBP 821 Section K. Section K addresses insurance documents. The H1 view carries the customs-value mapping; this paper does not claim an ISBP K1 freight/insurance test.

4.3 View C: DELTA IE H1 pack

The `delta_ie_h1_pack` view maps slots to UCC Annex B and DELTA IE H1 data elements (Table 3). The pack is `READY` only when every element is `COMPLETE`; `PENDING` means present but uncorroborated, `MISSING` means absent.

4.4 The Oracle Boundary Theorem

Theorem. Let $\mathrm{Seal}(s)$ denote a valid HMAC-SHA256 and AES-256-GCM seal on slot s under K_{slot}, let $\mathrm{Attest}(s)$ denote a valid Ed25519 attestation over the payload hash under a trusted attester key, and let $\mathrm{Physical}(s)$ denote that the physical event described by s occurred as stated. Then

$$\mathrm{Seal}(s) \not\Rightarrow \mathrm{Physical}(s), \qquad \mathrm{Attest}(s) \not\Rightarrow \mathrm{Physical}(s). \quad (22)$$

Proof sketch. $\mathrm{Seal}(s)$ is a function of the payload, the master secret, the dossier salt, the slot identifier, and the declared state. It is computed inside the tenant key domain and is invariant under the truth of $\mathrm{Physical}(s)$. $\mathrm{Attest}(s)$ proves that the holder of a trusted attester key signed the payload hash; it does not prove that the attester observed the event correctly, that the sample came from the declared batch, or that the accreditation is unrevoked. A tenant can seal a false payload; a trusted oracle can attest a false observation; a compromised device can report false GPS. Validity proves tenant-side integrity and attester authenticity, not physical truth.

Physical truth enters only through the `ASSERTED` $\rightarrow$ `CORROBORATED` transition, which requires an independent attester whose type is not `PARTY_SELF_DECLARATION`, a verified accreditation identifier (ISO/IEC 17020, ISO/IEC 17025, sovereign registry, carrier eBL registry), a valid Ed25519 signature over the payload hash, and a public key present in the relying party's trust store for that authority and slot scope at t_{ref}. The accreditation and trust-store layers establish competence and identity; the cryptographic layer secures the statement after capture and makes it independently verifiable. All are necessary; none is sufficient. OOES-18 verifies accreditation presence, key trust, and expiry, not real-time revocation; implementations SHOULD maintain a revocation feed. Oracle collusion, stolen attester keys, compromised tenants, and clock manipulation remain institutional threats outside the cryptographic boundary.

4.5 Conformance and availability

An implementation conforms to OOES-18 v1.1.0 if it accepts the Draft 2020-12 schema, reproduces every digest and Ed25519 signature in `vectors/CASCADE-2026-9F4A2C.dossier.json` and `trust/attesters.json` from the pinned fixtures, hard-stops both gates on the default vector with the exact slot sets, demotes a mutated `L3.S09` to `UNKNOWN` with `GATE_02` halt, rejects a fully corroborated dossier with a container, stuffing-geofence, origin-criterion, carrier, or CIF contradiction, rejects a valid signature from an untrusted key or a mutated signature, and computes identical blind indices for the two formatted bank inputs. The schema, vector, trust store, TypeScript verifier, and seventeen-test suite are released under Apache-2.0; this paper is CC-BY-4.0.

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